Day 5 Study Notes

Tue 17 Nov 2026 · Sprint 60 · JEE Main 2027

Part	Time	What to do
Study	3 h 30 min	Physics: Photoelectric effect and matter waves (75 min) · Maths: Vectors (part 2: cross product) (75 min) · Chemistry: d- and f-block elements (60 min)
Exam	1 h 15 min	The Day 5 exam in the Sprint 60 app: 30 questions, timed, no notes.
Review	1 h 15 min	Open every wrong and skipped question in the app, read its solution, tag the reason, then use Retry my mistakes.

How to use each chapter note. Read it once without stopping. Cover the formulas and write them from memory. Work the solved examples on paper before reading the working. Then read the NCERT section listed at the end.

Physics · Photoelectric effect and matter waves

About one question per paper, nearly always a substitution into Einstein's equation or the de Broglie formula. Learn the shortcut $E(\text{eV}) = \dfrac{1240}{\lambda(\text{nm})}$ and most questions take under a minute.

1. The photon

- Energy: $E = h\nu = \dfrac{hc}{\lambda}$. In practical units, $E(\text{eV}) = \dfrac{1240}{\lambda(\text{nm})}$.
- Momentum: $p = \dfrac{h}{\lambda} = \dfrac{E}{c}$. Rest mass is zero.
- A source of power P emits $N = \dfrac{P}{h\nu} = \dfrac{P\lambda}{hc}$ photons per second.
- Intensity is the energy per unit area per second. For a fixed frequency, more intensity means more photons, not more energetic photons.

2. What the experiments show

Observation	Depends on
Whether electrons are emitted at all	frequency only: it must exceed the threshold frequency ν_0
Number of electrons (saturation current)	intensity
Maximum kinetic energy and stopping potential	frequency, not intensity
Time delay	none: emission is instantaneous

3. Einstein's photoelectric equation

$$h\nu = \phi_0 + K_{\max}, \qquad K_{\max} = eV_0$$

- Work function: $\phi_0 = h\nu_0 = \dfrac{hc}{\lambda_0}$, where λ_0 is the threshold wavelength.
- Stopping potential: $V_0 = \dfrac{h}{e}\nu - \dfrac{\phi_0}{e}$.
- The graph of V_0 against ν is a straight line. Its slope is $\dfrac{h}{e}$, the same for every metal. It cuts the frequency axis at ν_0.
- In eV: $K_{\max}(\text{eV}) = \dfrac{1240}{\lambda(\text{nm})} - \phi_0(\text{eV})$, and the stopping potential in volts has the same number as $K_{\max}$ in eV.

> **Solved example 1.** Light of wavelength 400 nm falls on a metal of work function 2.0 eV. Find the maximum kinetic energy and the stopping potential.
> Photon energy $= \dfrac{1240}{400} = 3.1$ eV. $K_{\max} = 3.1 - 2.0 = 1.1$ eV, so $V_0 = 1.1$ V.

4. Matter waves

A particle of momentum p has a de Broglie wavelength:

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mqV}}$$

- Electron accelerated from rest through V volts: $\lambda = \dfrac{12.27}{\sqrt{V}}$ Å $= \dfrac{1.227}{\sqrt{V}}$ nm.
- Same kinetic energy: $\lambda \propto \dfrac{1}{\sqrt{m}}$. Same momentum: same wavelength. Same speed: $\lambda \propto \dfrac{1}{m}$.
- Same accelerating potential: $\lambda \propto \dfrac{1}{\sqrt{mq}}$.
- The Davisson–Germer experiment confirmed the wave nature of electrons by diffraction.

Solved example 2. A proton and an alpha particle have the same kinetic energy. Find the ratio of their de Broglie wavelengths.
$\lambda \propto \dfrac{1}{\sqrt{m}}$, so $\dfrac{\lambda_p}{\lambda_\alpha} = \sqrt{\dfrac{m_\alpha}{m_p}} = \sqrt{4} = 2$.

Traps. Doubling the intensity does not change the stopping potential. Doubling the frequency does not double $K_{\max}$, because the work function is subtracted first. In ratio questions, check which quantity is the same: energy, momentum, speed or accelerating voltage. For an alpha particle accelerated through a potential, the charge is $2e$.

Read in NCERT: Physics Class 12 Part 2, chapter "Dual Nature of Radiation and Matter".

Mathematics · Vectors (part 2: cross product)

The cross product supplies areas, perpendicular directions and the shortest-distance formula used in 3D geometry. The syllabus stops at the scalar and vector products of two vectors; triple products are not included.

1. Definition

$$\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin\theta\, \hat{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$\hat{n}$ is the unit vector perpendicular to both $\vec{a}$ and $\vec{b}$, given by the right-hand rule. In components: $\vec{a} \times \vec{b} = (a_2b_3 - a_3b_2,\ a_3b_1 - a_1b_3,\ a_1b_2 - a_2b_1)$.

2. Properties

- $\vec{b} \times \vec{a} = -\vec{a} \times \vec{b}$, and $\vec{a} \times \vec{a} = \vec{0}$.
- $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$.
- Non-zero vectors are parallel exactly when $\vec{a} \times \vec{b} = \vec{0}$.
- $\sin\theta = \dfrac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|}$.
- Lagrange's identity: $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2|\vec{b}|^2$.

3. Uses

Quantity	Formula
Area of a parallelogram with adjacent sides $\vec{a}$, $\vec{b}$	$\lvert\vec{a} \times \vec{b}\rvert$
Area of a triangle with sides $\vec{a}$, $\vec{b}$	$\frac{1}{2}\lvert\vec{a} \times \vec{b}\rvert$
Area of triangle ABC	$\frac{1}{2}\lvert\vec{AB} \times \vec{AC}\rvert$
Area of a parallelogram with diagonals $\vec{d}_1$, $\vec{d}_2$	$\frac{1}{2}\lvert\vec{d}_1 \times \vec{d}_2\rvert$
Unit vectors perpendicular to both $\vec{a}$ and $\vec{b}$	$\pm\dfrac{\vec{a} \times \vec{b}}{\lvert\vec{a} \times \vec{b}\rvert}$
Moment of a force about a point	$\vec{r} \times \vec{F}$

Three points A, B, C are collinear when $\vec{AB} \times \vec{AC} = \vec{0}$.

> **Solved example 1.** Find the area of the triangle with vertices $A(1,1,2)$, $B(2,3,5)$, $C(1,5,5)$.
> $\vec{AB} = (1,2,3)$ and $\vec{AC} = (0,4,3)$. $\vec{AB} \times \vec{AC} = (2\cdot3 - 3\cdot4,\ 3\cdot0 - 1\cdot3,\ 1\cdot4 - 2\cdot0) = (-6,-3,4)$, with magnitude $\sqrt{61}$. Area $= \dfrac{\sqrt{61}}{2}$.

> **Solved example 2.** $|\vec{a}| = 2$, $|\vec{b}| = 5$ and $|\vec{a} \times \vec{b}| = 8$. Find $|\vec{a} \cdot \vec{b}|$.
> By Lagrange's identity, $(\vec{a} \cdot \vec{b})^2 = 4 \times 25 - 64 = 36$, so $|\vec{a} \cdot \vec{b}| = 6$.

Traps. The middle component of the determinant carries a minus sign; writing the components in the cyclic form above avoids the mistake. With diagonals, remember the factor of one half. $\vec{a} \times \vec{b} = \vec{a} \times \vec{c}$ does not give $\vec{b} = \vec{c}$; it gives $\vec{a}$ parallel to $\vec{b} - \vec{c}$. If both $\vec{a} \cdot \vec{b} = 0$ and $\vec{a} \times \vec{b} = \vec{0}$, at least one of the vectors is zero.

Read in NCERT: Mathematics Class 12 Part 2, chapter "Vector Algebra": the vector (cross) product, with the solved examples and the miscellaneous exercise.

Chemistry · d- and f-block elements

One to two questions per paper. The favourites are the reactions of $KMnO_4$ and $K_2Cr_2O_7$, magnetic moments, coloured ions and lanthanoid contraction.

1. Electronic configuration

- General: $(n-1)d^{1-10}\ ns^{1-2}$. Exceptions in the 3d series: Cr is $3d^5 4s^1$ and Cu is $3d^{10}4s^1$.
- When ions form, the 4s electrons are removed before the 3d electrons. Fe^{2+} is $3d^6$, Fe^{3+} is $3d^5$.
- Zn, Cd and Hg have full d^{10} shells in both atoms and ions, so they are not counted as transition elements.

2. Trends in the 3d series

- Atomic radius falls at first, stays nearly constant in the middle, and rises slightly at the end.
- Melting points and enthalpies of atomisation are high because d electrons take part in metallic bonding. Mn is unusually low; Zn is lowest.
- **Oxidation states.** Mn shows the largest number, up to +7. The highest states appear with oxygen and fluorine. Sc shows only +3 and Zn only +2.
- Mn^{2+} (d^5) and Zn^{2+} (d^{10}) are especially stable. Cr^{2+} is a reducing agent; Mn^{3+} is an oxidising agent.
- Copper has a positive $E^\circ(\mathrm{Cu}^{2+}/\mathrm{Cu})$, so it does not release hydrogen from acids.
- Cu^+ is unstable in water and disproportionates: $2Cu^+ \rightarrow Cu^{2+} + Cu$. The much larger hydration enthalpy of Cu^{2+} more than pays for the second ionisation.

3. Magnetism and colour

$\mu = \sqrt{n(n+2)}$ BM, where n is the number of unpaired electrons.

Ion	Configuration	Unpaired	Colour in water
Sc^{3+}	d^0	0	colourless
Ti^{3+}	d^1	1	purple
V^{3+}	d^2	2	green
Cr^{3+}	d^3	3	violet
Mn^{2+}	d^5	5	pale pink
Fe^{2+}	d^6	4	green
Fe^{3+}	d^5	5	yellow
Co^{2+}	d^7	3	pink
Ni^{2+}	d^8	2	green
Cu^{2+}	d^9	1	blue
Zn^{2+}	d^{10}	0	colourless

4. Other characteristic properties

- **Catalysts:** V_2O_5 in the contact process, iron in the Haber process, nickel in hydrogenation, MnO_2 in the decomposition of $KClO_3$.
- **Interstitial compounds** (small atoms such as H, C, N in the lattice) are hard, have high melting points and keep metallic conductivity.
- **Alloys:** brass is Cu–Zn; bronze is Cu–Sn.

5. Potassium dichromate, $K_2Cr_2O_7$

- Made from chromite ore: $4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$. The chromate is acidified to dichromate and then treated with KCl.
- Yellow chromate and orange dichromate interconvert with pH: $2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$. Chromium is +6 in both.
- Oxidising action in acid: $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$. One mole takes 6 moles of electrons, so it oxidises 6 moles of Fe^{2+} or 6 moles of I^-.
- Chromate ion is tetrahedral. Dichromate is two tetrahedra sharing one oxygen corner.

6. Potassium permanganate, $KMnO_4$

- Made from pyrolusite: $2MnO_2 + 4KOH + O_2 \rightarrow 2K_2MnO_4$ (green manganate) $+ 2H_2O$. The manganate is then oxidised, or disproportionates in acid, to purple permanganate.

Medium	Half reaction	Electrons gained
Acidic	$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$	5
Neutral or faintly alkaline	$MnO_4^- + 2H_2O + 3e^- \rightarrow MnO_2 + 4OH^-$	3
Strongly alkaline	$MnO_4^- + e^- \rightarrow MnO_4^{2-}$	1

- In acid it oxidises Fe^{2+} to Fe^{3+}, oxalate to CO_2, I^- to I_2, nitrite to nitrate and H_2S to sulphur.
- With oxalate: $2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$.
- In neutral or alkaline solution it oxidises I^- to IO_3^- and thiosulphate to sulphate.
- Hydrochloric acid is not used to acidify it, because permanganate oxidises HCl to chlorine.
- Permanganate is diamagnetic; manganate has one unpaired electron. Both ions are tetrahedral.

7. Lanthanoids and actinoids

- Lanthanoids fill the 4f orbitals. The common oxidation state is +3. Ce^{4+} (f^0) is an oxidising agent; Eu^{2+} (f^7) and Yb^{2+} (f^{14}) are reducing agents.
- **Lanthanoid contraction:** the steady fall in size from La to Lu, caused by the poor shielding of 4f electrons.
- Its consequences: Zr and Hf have almost the same radius; lanthanoids are hard to separate; the basic strength of the hydroxides falls from $La(OH)_3$ to $Lu(OH)_3$.
- Actinoids fill the 5f orbitals, are radioactive, and show a wider range of oxidation states (up to +7) because the 5f, 6d and 7s levels are close in energy.

Traps. Count d electrons from the ion, not the atom. Mn^{2+} and Fe^{3+} both have five unpaired electrons. The product of permanganate depends on the medium: Mn^{2+} in acid, MnO_2 in neutral solution, manganate in strong alkali. In chromate–dichromate interconversion the oxidation state does not change.

Read in NCERT: Chemistry Class 12 Part 1, chapter "The d- and f-Block Elements". Read the sections on $K_2Cr_2O_7$ and $KMnO_4$ line by line.